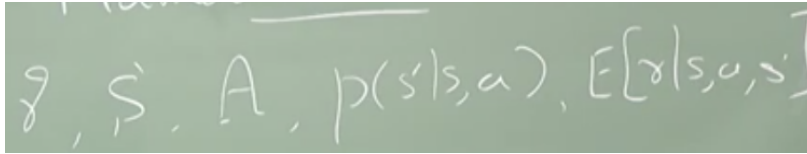


w4L21 MDP Modeling (33min)



MDP specified by S , A , transition prob and $E[r|s, a, s']$, and possibly gamma

- RL books - they will leave out gamma
- They don't acknowledge that gamma changes the solution

Recycling Robot:

- Finds all empty cans - puts in recycling system
- It will have limited battery power - how it spends battery
- If going to run out of battery - recharge, sit in a place near recycle bin
 - Can use it like a trash can
- If runs out of battery - need to carry robot to charging station
 - Don't want it to sit idle

What are the states, actions, rewards?

1. Rewards: easy ->

- gathers can, give it a reward of +1
- Don't run out of battery -> say give -3 (not that bad)
 - After collecting 15 cans, dies, not bad
- If chooses to charge -> give positive reward? = 0
 - No reward makes sense
 - Just for get reward it will recharge
 - You can get more rewards in future
 - immediate reward it looks bad - for future it is attractive
- Battery level - How detailed do you want to make it
 - Continuous
 - Discretisation

2. States:

Charge - treat as a number - can do Low, Medium, High

- Only when it is low, you will charge, so can make it low and High
- Recharge -> i will have enough battery to go to recharge point

Will not care: No. of cans

How many cans - it has

Details taken care: Pose of robot, velocity of robot

Location of charging point is nearby - as soon as battery is low

Giving word problem and asking to convert to MDP

- More constrained problem
- Can make lot of choices you make

Designing RL problem

- State, Action, Reward, Transition probability
- In future explicitly may not need transition probability

3. Actions:

- Actively seek cans
- Wait for cans
- Recharge

From low can to high with penalty of -3 (equivalent to zero charge state)

- If high is 100% - after action will not go to low
- If 40% is cutoff - after taking action at 40% will go to low
- Sometimes it goes from high to low, sometime it will remain in high after action

If you know everything about the world

- It is not stochastic
- It is better to give up information - what will be the right place to draw line

Have to relate Rewards to State and Action?

- Actively seeking cans - prob i will find a can - need to specify
- If waiting for cans - what is the probability

I can act as dustbin - 12:30 to 1:30, will get lot of cans - i will not seek - just sit

- After 1:30 i may have to go and seek the cans

What could be optimal policy

- You are designing the problem it self depending upon what you think the solution should be
- Which is bad think
- **Let the Algorithm find Solution**

Is there anything I need to know

- If I am actively seeking what is the prob i will run out of battery
- When i am sitting what is the prob i will run out of battery

If I am in High state- actively seeking cans - state matters - will go to state high

If I am in High state- actively seeking cans - state matters - will go to state low

If I am in Low state- actively seeking cans - state matters - will go to state low

Prohibit - recharge action in state high

$$\gamma_{can} = +1$$

$$\gamma_{out. of battery} = -3$$

$$\gamma_{recharge} = 0$$

State:
 Charge: Low, High.
 X No of cans - - -

Actions:
 - Actively seek cans, α
 - Wait for cans, β $\alpha > \beta$
 - Recharge

alpha will be higher than beta

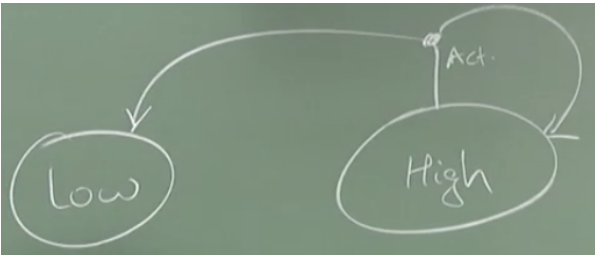
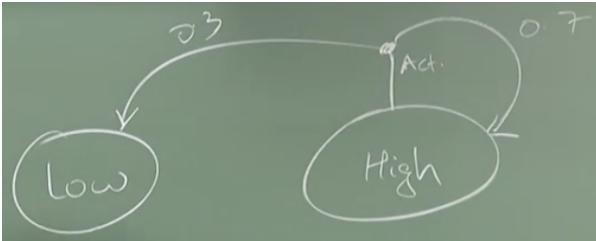
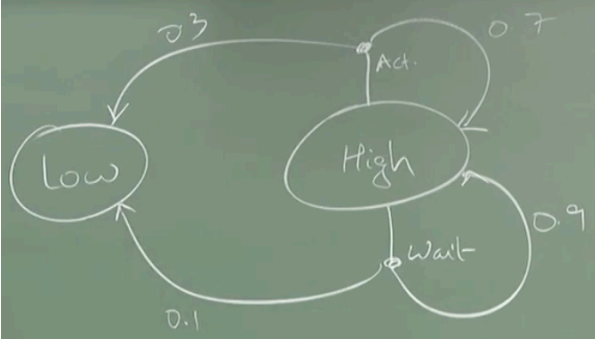
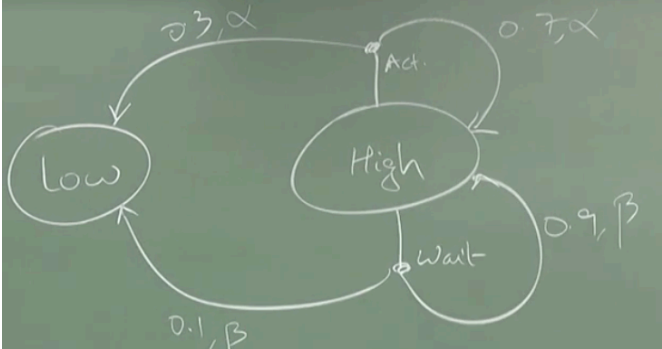
- We have not modelled time

$$p(s'|s, a),$$

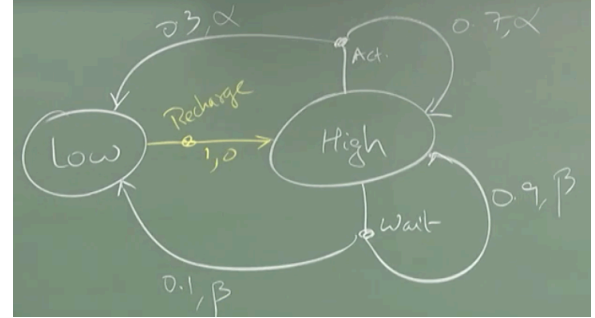
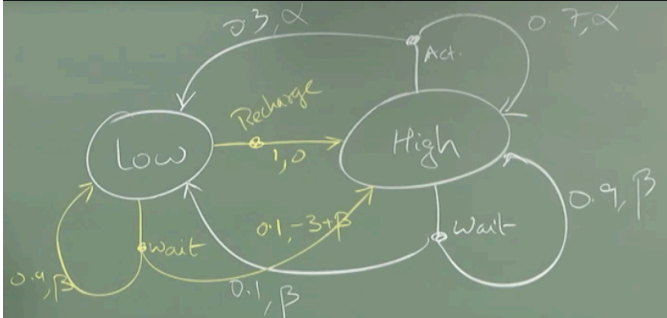
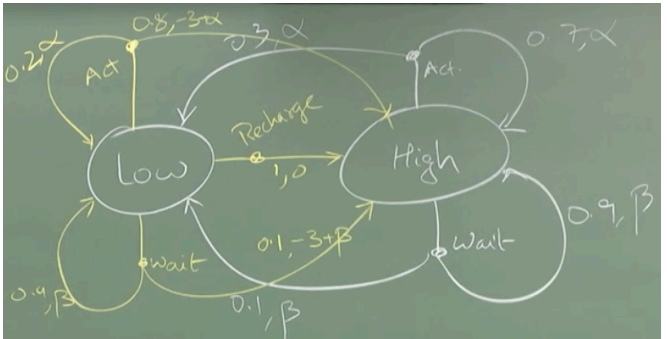
We are NOT talking about SOLUTION

- We are defining problem

You are giving state sufficientness

	Add some prob 
Other action 	What the reward is going to be Expected reward is alpha 

From - low 3 actions

Recharge - will go to high with 0 reward 	Wait: 
Simple problem We chose to ignore lot of factors Took time to come up with - Simple trackable MDP	Actively seeking 

w4L22 Bellman Equation (14min)

Value function:

$$V^\pi(s) = E_\pi[G_t | S_t = s]$$
$$Q^\pi(s, a) = E_\pi[G_t | S_t = s, A_t = a]$$

v-pi-s starting in state s following policy pi there after - taking expectations of the returns

Unroll expectations

$$V^\pi(s) = E_\pi[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s]$$

$$V^\pi(s) = E_\pi[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s]$$
$$= E_\pi[R_{t+1} + \gamma V^\pi(s') | S_t = s]$$

Expected value of G_{t+1} -> whatever state you land up -> G_t + one term

$$V^\pi(s) = E_\pi[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s]$$
$$= E_\pi[R_{t+1} + \gamma V^\pi(s') | S_t = s]$$

It is partial expectation because s' can change

So I need to condition on s' also - for full expectations take out

Think as generative process - start in state s , select action according to π policy,

$$\begin{aligned} v^\pi(s) &= E_\pi \left[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \mid s_t = s \right] \\ &= E_\pi \left[R_{t+1} + \gamma v^\pi(s') \mid s_t = s \right] \\ &= \sum_a \pi(a|s) \sum_{s'} p(s'|s,a) \left\{ E[r|s,a,s'] + \gamma v^\pi(s') \right\} \end{aligned}$$

Taken action a according to π ,
determined s' - what the next state is going to be s' then determine $E[r \mid s, a]$, as well as $v^\pi(s')$

$$\begin{aligned} v^\pi(s) &= E_\pi \left[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \mid s_t = s \right] \\ &= E_\pi \left[R_{t+1} + \gamma v^\pi(s') \mid s_t = s \right] \\ v^\pi(s) &= \sum_a \pi(a|s) \sum_{s'} p(s'|s,a) \left\{ E[r|s,a,s'] + \gamma v^\pi(s') \right\} \end{aligned}$$

It is Bellman's equation for v^π

Whole thing gives me - Expected return - do it over all possible outcomes

- Expected outcome for choice of a and s' (summed over a and s')

$E_\pi[R_{t+1}]$ is given in MDP

We have written v^π in terms of v^π

It is system of equation - one equation for every state

- No. of variable = no. of states
- Variable is $v^\pi(s)$
- N such eq and N such variables - when can you not solve it
 - When you have dependent column
 - HAVE UNIQUE solution
 - P is stochastic - rows sum to 1

It is Bellman's equation for q-pi

- Think of it in generative process
- Start with s and a is picked $\pi(a|s)$ is out of picture

$$q_v^\pi(s, a) = \sum_{s'} p(s'|s, a) \left\{ E[r|s, a, s'] + \gamma \right\}$$

I could write v-pi

I will not write q-pi in terms of q-pi

$$q_v^\pi(s, a) = \sum_{s'} p(s'|s, a) \left\{ E[r|s, a, s'] + \gamma \sum_{a'} \pi(a'|s') q_v^\pi(s', a') \right\}$$

One more ancillary equation

$$v^\pi(s) = \sum_a \pi(a|s) \cdot q_v^\pi(s, a)$$

That's how q and v are related

Can't write $q_v^\pi(s, a)$ in terms of $v^\pi(s)$

- It has to be v^π (of subsequent state)

w4L23 Bellman Optimality Equation (30min)

Optimality - want π , so no other π can get better return than this π

$$\max_{\pi} E\{R_t\}$$

Expectation of over return doesn't make sense:

$$\max_{\pi} E_{\pi}\{G_t | S_t = s\} \quad \forall s$$

$$\max_{\pi} V^{\pi}(s) \quad \forall s$$

Do it pointwise:

$$\pi^* = \arg \max_{\pi} V^{\pi}(s) \quad \forall s$$

There is no THE optimal policy, there could be multiple optimal policy

Can I go up, down, left, right - every action will get reward -1

- Stochastic prob 0.9 go up, 0.1 can't go up (may be wheel of robot stuck/goes back)
- 12 states - can choose any of the two actions
- It is episodic task, will end when you reach G
- Huge number of optimal policy
- In deterministic, can take how many steps will it take

		-

Is there anything else common in the optimal policy

- Regardless of policy I chose, value of the state will be same

$$\pi^* = \arg \max_{\pi} V^{\pi}(s) \quad \forall s$$
$$V^{\pi^*} \triangleq V^* = \max_{\pi} V^{\pi}(s) \quad \forall s$$

π^* need not be unique, V^* will be unique (optimal value function)

$$\underbrace{\pi^*}_{\text{optimal policy}} = \arg \max_{\pi} V^{\pi}(s) \quad \forall s$$
$$V^{\pi^*} \triangleq V^* = \max_{\pi} V^{\pi}(s) \quad \forall s$$

↑
Optimal value fn.

How will same π reach $V(s)$

- Finite - easy to show

Next class - why an optimal policy exists

Optimal Value function for q

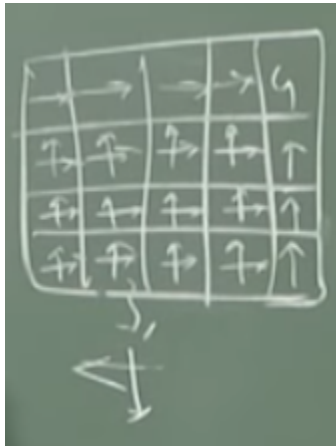
$$Q^*(s, a) = \max_{\pi} \sum_{s'} p(s' | s, a) \cdot [E[r | s, a, s'] + \gamma V^{\pi}(s')]$$

Even though i am talking about optimal policy - 1st action i am taking action a
After i take a, behave optimally

Max can be inserted

$$q^*(s, a) = \sum_{s'} p(s' | s, a) \cdot [E\{r | s, a, s'\} + \gamma \max_{\pi} V^{\pi}(s')]$$

Anything before is not influenced by max, as it is fixed (a is fixed - remember)



for each state q^* can be defined even for sub-optimal actions

What will be $v^*(s)$ in terms of q^* ?

$$q^*(s, a) = \sum_{s'} p(s' | s, a) \cdot [E\{r | s, a, s'\} + \gamma \max_{\pi} V^{\pi}(s')]$$

q^* tells - if i take action a and behave optimally there after

$$V^*(s) = \max_a q^*(s, a)$$

v^* tells me - behave optimally from current state - taking max, which action gives best

- Finding what optimal value function
- What is action a, i check immediate and further - optimal

$$\begin{aligned}
 V^*(s) &= \max_a q^*(s, a) \\
 &= \max_a \sum_{s'} p(s'|s, a) [E\{r|s, a, s'\} + \gamma V^*(s')]
 \end{aligned}$$

Max over v-pi is v*

This is Bellman optimality equation for v*

Can write Bellman optimality equation for q*

As long as i have policy - assigns non-zero prob - is still optimal policy

- .5 prob go up, down
- There are infinite number of policies

Given a value function (v*), how will you recover optimal policy from there?

- What action i have to take in each state
- Take one which gives max over it

Utility for taking an action:

$$\sum_{s'} p(s'|s, a) [E\{r|s, a, s'\} + \gamma V^*(s')]$$

Pick action which gives pi*(s):

$$\pi^*(s) = \arg \max_a \sum_{s'} p(s'|s, a) [E\{r|s, a, s'\} + \gamma V^*(s')]$$

Argmax - one maximizer or all maximizer (distribution - uniform or other)

- Still need to know p and other para to recover, Expectation

Given q function, how will you recover optimal policy from there?

Do max of

$$\pi^*(s) = \arg \max_a q^*(s, a)$$

$$q^*(s, a) = \sum_{s'} p(s'|s, a) \cdot [E\{r|s, a, s'\} + \gamma \max_{a'} q^*(s', a')]$$

- I can be greedy wrt q^* and get π^*
- Don't have to do one step look ahead (like v^*)

Talked about solve Bellman for v-pi and q-pi

Can you solve for - N eq and N var

It is not linear because of max: - nonlinear system of eq

$$v^*(s) = \max_a \sum_{s'} p(s'|s, a) [E\{r|s, a, s'\} + \gamma v^*(s')]$$

Next class: Show v-pi and v^* are unique - multiple ways of solving for v^* and q^*

Rest of the course ->

- multiple ways of solving for v^* and q^*
- Depending upon how much information you have

ToDo: w4L24 Cauchy Sequence and Greens Eq.
(31min)

ToDo: w4L25 Banach FixedPoint Theorem (25min)